TATHAGATA NATH

AI / Agentic AI | Machine Learning | Deep Learning

My Online Resume

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PROFESSIONAL SUMMARY

Computer Science professional with an M.Tech in Computer Science & Engineering and hands-on project experience in Agentic AI, Machine Learning, Deep Learning, and Data Analysis . Skilled in Python, LangChain, and LangGraph, NumPy, Pandas, SQL, ANN/CNN, ML(Regression and Classification) with experience developing prediction systems and AI-powered applications. Published research in wireless sensor networks and experienced in web design and scientific review activities. I have more than 5 years year gap but if you give me a chance then I can prove myself.

TECHNICAL SKILLS

Generative / Agentic AI	LangChain, LangGraph, Groq-based AI applications, custom tools
Programming	Python, NumPy, Pandas, SQL
Data Science	Data Preprocessing, Data Analysis, Data Analytics
Machine Learning	Machine Learning(Regressor and Clasification), Deep Learning, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN)
Other	MATLAB, Database Management Systems, Data Structures

INTERNSHIP EXPERIENCE

Website Designer **Jun 2016 – Nov 2017**

Creazione Production, Kolkata

- Worked on website design and related web-development activities for the organization using wordpress.

EDUCATION

M.Tech, Computer Science & Engineering — Aliah University, Kolkata, West Bengal **CGPA: 8.63 | 2019**

5-Year Integrated M.Sc, Computer Science — Aliah University, Kolkata, West Bengal **CGPA: 7.67 | 2016**

Higher Secondary — Manbazar R.M.I, WBCHSE **60% | 2010**

Secondary — Manbazar R.M.I, WBBSE **74.63% | 2007**

PROJECTS

0.1 LANGGRAPH PROJECTS

1. AI Used-Car Price & Purchase Advisor | Streamlit | LangGraph | Groq LLM | FastAPI | Machine Learning Regression | DuckDuckGo

Developed an AI-powered used-car price and purchase recommendation system combining Machine Learning regression with Generative AI and web research. Built a regression pipeline to estimate vehicle prices using attributes including brand, age, mileage, engine capacity, fuel type, transmission, ownership history, and location. Deployed the ML model as a FastAPI REST API and implemented a LangGraph workflow to orchestrate input understanding, price prediction, web research, and final AI recommendations. Integrated Groq LLM and DuckDuckGo Search to analyze vehicle reliability, common problems, maintenance considerations, and ownership risks. Developed an interactive Streamlit application providing estimated price and AI-driven BUY / CONSIDER / AVOID recommendations. **Live Demo: *Used-Car Advisor***, **Code: *GitHub***

2. Cardiovascular Disease Prediction AI Medical Research Assistant1. Technologies: ML classification, Scikit-learn, FastAPI, Streamlit, LangGraph, Groq LLM, DuckDuckGo Search, SQLite

Developed an end-to-end machine learning application to predict categories of cardiovascular disease using structured patient health data.

Built a LangGraph workflow to orchestrate prediction, BMI calculation, medical information retrieval, and AI-generated health information.

Integrated a Groq-hosted large language model and DuckDuckGo search to retrieve relevant medical information and support evidence-aware responses.

Designed the application to distinguish model predictions from clinical diagnoses and medically validated risk assessments. **Live Demo: *Medical Report***, **Code: *GitHub***

3. AI Multi-Agent Travel Planner using LangGraph and LLM

Built an AI-powered travel planning system using LangGraph and LLM orchestration to generate end-to-end travel itineraries. Integrated real-time flight data (AviationStack API) and hotel recommendations (Tavily search) to provide contextual travel suggestions. Developed an interactive Streamlit web app enabling users to input travel requests and receive structured, AI-generated travel plans. **Live Demo: *Travel Planner***, **Code: *GitHub***

0.2 LANGCHAIN PROJECTS

1. AI-Powered Live Web Search Assistant using Groq LLM and LangChain
Built a real-time AI question answering system using Groq LLM, LangChain, and DuckDuckGo search to retrieve and process up-to-date web information. Developed an interactive Streamlite web application enabling users to ask open-domain questions and receive concise, context-aware answers instantly. **Live Demo:** ***Search Assistant***,

Code: ***GitHub***

2. LLM-Powered Multi-Agent Stock and Mutual Fund Analysis System
Built a real-time AI question answering system using Groq LLM, LangChain, and DuckDuckGo search to retrieve and process up-to-date web information. Developed an interactive Streamlite web application enabling users to ask open-domain questions and receive concise, context-aware answers instantly. **Live Demo:** ***Search Assistant***,

Code: ***GitHub***

3. AI powered City information and Weather update Assistant | LangChain, Qwen3-32B, Streamlit
* Built an **Agentic AI Weather and City information assistant** using LangChain and Qwen3-32B with tools for real-time weather, astronomy, and location research. * Integrated WeatherAPI, DuckDuckGo Search, and SQLite checkpointing for **live data retrieval and persistent agent state**. * Developed and deployed an interactive **Streamlit application** with secure API-key management using Streamlit Secrets.

Live Demo: ***Search Assistant***, **Code:** ***GitHub***

4.AI Resume Analyzer | Python, LangChain, Groq LLM, Streamlit, PyPDF
* **AI Resume Screening ATS Analyzer** | Python, LangChain, Groq LLM, Streamlit, PyPDF
* Built an LLM-powered ATS resume screening system that analyzes PDF resumes and generates structured candidate insights. * Designed a LangChain-based pipeline for PDF ingestion, text extraction, prompt orchestration, and AI-driven resume evaluation. * Implemented structured JSON generation for extracting skills, education, experience, projects, skill gaps, and recruiter recommendations from unstructured resumes. * Developed a Streamlit web application enabling users to upload resumes and receive automated HR-style analysis and actionable career recommendations. * Applied prompt engineering techniques to improve output consistency, prevent information hallucination, and enforce schema-based responses. .

Live Demo: ***Analyze Resume***, **Code:** ***GitHub***

5.AI-Powered RAG Application | LangChain, Streamlit, Hugging Face | Technologies: Python, Streamlit, LangChain, Hugging Face, FAISS, BGE Embeddings, PyPDF, RAG
Developed an interactive Retrieval-Augmented Generation (RAG) application to answer user queries using information retrieved from uploaded PDF documents and website URLs.
Implemented document ingestion and text chunking using PyPDFLoader, WebBaseLoader, and RecursiveCharacterTextSplitter.
Generated semantic embeddings using the BAAI/bge-small-en-v1.5 model and stored document vectors in a FAISS vector database.
Built a similarity-based retrieval to retrieve the top 4 relevant document chunks for each query.
Integrated a Hugging Face-hosted LLM through LangChain to generate context-grounded answers using custom prompt templates.
Created an interactive Streamlit interface supporting multiple PDF uploads, website ingestion, knowledge-base creation, and question answering. .

Code: ***GitHub***

0.3 DEEP LEARNING PROJECTS(CNN)

1. Diabetes Prediction using CNN
Deep learning using CNN for retinal image detection **Code:** ***GitHub***

0.4 MACHINE LEARNING PROJECTS

1. Bengaluru House Price Prediction
ML regression model for house price prediction ***HOUSE PRICE PREDICTION***,

Breast Cancer Prediction using Python
Machine learning project for breast-cancer prediction using Python. **Code:** ***GitHub***

0.5 ACADEMIC PROJECTS

Wireless Sensor Network for Rhino Protection
M.Tech research project on applying wireless sensor networks for rhino monitoring against poaching in Kaziranga National Park. **Technology:** MATLAB. **Project Paper:** *Download*

Urdu Offline Signature Identification
M.Sc research project on offline Urdu signature identification. **Technology:** MATLAB

RESEARCH & PUBLICATIONS

- Tathagata Nath, Md Azharuddin, "Application of wireless sensor networks for Rhino protection against poachers in Kaziranga National Park," AEU-International Journal of Electronics and Communications (Elsevier), Vol. 111, 2019.***(SCI Journal impact factor 3.1)***. **Research Paper:** *Download*

ACHIEVEMENTS & PROFESSIONAL ACTIVITIES

- Best Startup Idea: "Automatic Postal Kiosk," International Conference on Frontiers in Computing and Systems, ***(Springer)***. **Certificate:** *Download*.
- Contributed as a reviewer for the International Journal of Electrical and Computing Engineering ***(Scopus)***. **Certificate:** *Download*.

CORE ACADEMIC SUBJECTS

Data Structures | DBMS

Certificates

1.Foundation: Introduction to Langchain-python. Langchain Certification from Langchain academy.
(Show Certificate)
2.Introduction to Langgraph-python. Langgraph Certification from Langchain academy.
(Show Certificate)
3.Completed Machine Learning training and practical projects using Python. from Udemy
(Show Certificate)
4.What is data Science Course. authorized by IBM and offered by Coursera.
(Show Certificate)

Extra- Curriculum activities

Bengali story writer : *Read my Story*

ADDITIONAL INFORMATION

Career Status: Returning to the professional workforce and seeking opportunities to apply and further develop skills in AI, Machine Learning, Deep Learning, and Data Science.